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JP2012002736A 20120105

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SUBJECT OF THE INVENTION

TECHNICAL FIELD

[**0001**] This invention relates to an exhaust_gas|exhaustion RI density|concentration reduction apparatus.

DESCRIPTION OF RELATED ART

[**0002**] It is as follows, in order to administer the radioactive tracer which consists of a radioisotope (radioisotope; Henceforth "RI") to a test subject so that it may describe in the following patent document 1 conventionally,
The positron emission tomography (Positron Emission Tomography; Henceforth "PET") apparatus which can perform efficiently preparation and delivery of the said radioactive tracer is known.

[**0003**] In such a PET apparatus, nitrogen and the mass number of 13 are used [the oxygen (The following, SP>13>N#v,<txf:notrans>SP>15>O#v,Ì,æ,¤,É•\,·) of 15, etc.], for example for a mass number as a radioactive tracer.
For example, RI gas called in <15>O, they are <15>O2 gas, C<15>O gas, or C<15>O2 gas etc. is used as a radioactive tracer.

PRIOR ART LITERATURE

[**0004**] Unexamined-Japanese-Patent No. 2005-324007

PROBLEM TO BE SOLVED BY THE INVENTION

[**0005**] The test|inspection by the above PET apparatuses is carried out in the laboratory which is a control area for radiation.
On the other hand, as for the administration to the test subject of RI gas, it is common to be carried out through the mask etc. which were attached to a test subject's opening|mouth,
In the case of this administration, some RI gas may leak out in a laboratory.
Usually, although a ventilation installation is provided in a laboratory, even if it is a case where RI gas leaks out in a laboratory, as for the exhaust gas discharged|emitted by the ventilation installation from a laboratory, RI density|concentration needs to be made into less than predetermined value in the eye of the law.

[0006] However, in the PET apparatus described in the said patent document 1, when leaking out of RI gas in a laboratory arose in this way, it did not clarify about the countermeasure against RI density|concentration in the exhaust gas in.
Therefore, there existed a possibility that RI density|concentration in exhaust gas might not be reduced desired.

[0007] Then, this invention is made in view of such a situation, and an object of this invention is to provide the exhaust_gas|exhaustion RI density|concentration reduction apparatus which can reduce RI density|concentration in exhaust gas desired.

MEANS TO SOLVE THE PROBLEM

[0008] The exhaust_gas|exhaustion RI density|concentration reduction apparatus concerning this invention is provided in the laboratory in which the PET apparatus was installed, and a discharge|emission delay means to delay outdoor discharge|emission of gas is provided, without stopping|fastening the flow of the gas discharged|emitted from the inside of a laboratory outdoor,
It is characterized by the above-mentioned.

[0009] Since according to the exhaust_gas|exhaustion RI density|concentration reduction apparatus concerning this invention outdoor discharge|emission of gas is delayed by the discharge|emission delay means even if it is a case where RI gas leaks out in a laboratory, the radioactivity of RI gas can be attenuated by this delay.
Therefore, RI density|concentration in exhaust gas can be reduced desired.
Moreover, since the flow of gas is not stopped|fastened, it is not necessary to provide the space for retention of gas, and gas can be discharged|emitted continuously.

[0010] Here, a discharge|emission delay means has a tubular part which lets gas pass,
If a tubular part has a curved part for changing the direction to which a tubular part extends, it will use few installation spaces effectively,
A longer pipe line can be provided and the reduction effect of RI density|concentration by the above-mentioned radioactive attenuation|damping can be exhibited suitably.

[0011] Moreover, since the discharge|emission delay means can change the flow direction of gas with a 1st partition plate if it has a 1st partition plate for changing the flow direction of gas,
The reduction effect of RI density|concentration by the radioactive attenuation|damping which could delay discharge|emission of gas reliably and described it above in the limited space can be exhibited suitably.
Moreover, the said effect can be exhibited by simple structure, without requiring complicated components.

[0012] Moreover, a discharge|emission delay means comprises the chamber which has the predetermined volume,
If the 2nd partition plate is provided in the chamber, it can prevent that gas short-circuits and flows into a chamber with a 2nd partition plate, and discharge|emission of gas can be delayed reliably.

[0013] Moreover, a laboratory has the 2nd division whose 1st division which is space in which a test subject is accommodated and 1st division are the partitioned-off space,
It can prevent that a discharge|emission delay means will expose it to the 1st division which accommodates a test subject if the discharge|emission delay means is arrange|positioned in 2nd division, and the improvement of an external appearance is achieved.

ADVANTAGE OF THE INVENTION

[0014] According to this invention, RI density|concentration in exhaust gas can be reduced desired.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015]

FIG. 1 :It is a schematic diagram of the PET system by which the exhaust_gas|exhaustion RI density|concentration reduction apparatus concerning 1st Embodiment of this invention is applied.
FIG. 2 :It is a schematic sectional drawing which shows the exhaust_gas|exhaustion RI density|concentration reduction apparatus in FIG. 1.
FIG. 3 :It is a schematic sectional drawing which shows the exhaust_gas|exhaustion RI density|concentration reduction apparatus concerning 2nd Embodiment.
FIG. 4 :It is a front view which shows schematic structure of the exhaust_gas|exhaustion RI density|concentration reduction apparatus concerning 3rd Embodiment.

EMBODIMENT OF THE INVENTION

[0016] Hereafter, embodiment of the exhaust_gas|exhaustion RI density|concentration reduction apparatus concerning this invention is described, referring drawings.
In addition, in description of drawing, the same code|symbol is attached|subjected to the same element, and the overlapping description is abbreviate|omitted.

[**0017**] (1st Embodiment)
FIG. 1 is a schematic diagram of the PET system by which the exhaust_gas|exhaustion RI density| concentration reduction apparatus concerning 1st Embodiment of this invention is applied.
It is for installing the PET system M shown in FIG. 1, for example in a medical institution etc., administering the radioactive tracer which changes from RI to test subject SB, and test|inspecting test subject SB by PET method.
This embodiment demonstrates the case where RI gas (<15>O2 gas, C<15>O gas, or C<15>O2 gas) which consists of <15>O is used as a radioactive tracer.
Test subject SB is made to suck|inhale the gas of a state by which RI gas, air, or oxygen gas was mixed in the PET test|inspection of this embodiment.
For example, the oxygen metabolism in a brain, the amount of regional blood flow, and the data of a blood pool image are acquired.

[**0018**] As shown in FIG. 1, the PET system M is provided with three chamber|rooms which have each different space.
Namely, the cyclotron chamber C for a cyclotron (not shown) to be provided and for the PET system M synthesize|combine RI gas.
While administering RI gas by which purity was measured in the hot lab chamber H for measuring the purity of RI gas synthesize|combined in the cyclotron chamber C, and the hot lab chamber H to test subject SB, the PET laboratory (laboratory) R for imaging the inside of the body of test subject SB, and carrying out a PET test|inspection is provided.
The equipment is provided in these three chamber|rooms,
This equipment is mutually connected by piping (henceforth a "line") construct|laid among chamber|rooms.
In addition, each of these chamber|room is segment|divided with the radiation-shielding wall 80 which prevents permeation|transmission of a radiation.
The radiation-shielding wall 80 can be formed, for example with concrete.

[**0019**] In addition, the cylinder 21 which stores the source gas for RI manufacture in the cyclotron chamber C is provided in outdoor [of the cyclotron chamber C].
Hereafter, the equipment with which the PET system M is provided in each room is demonstrated.

[0020] The target box 22 which irradiates a proton etc. with a cyclotron and radioactivates the source gas which is connected to the cylinder 21 by the line L1, and is supplied from the cylinder 21 in the cyclotron chamber C,
The line L2 connects in the secondary side of the target box 22, and the automatic-synthesis apparatus 23 which refine|purifies the radioactivated source gas which is supplied from the target box 22, and is made into RI gas is provided.
The automatic-synthesis apparatus 23 sends out synthetic|combination RI gas to the hot lab chamber H through the line L3.

[0021] In addition, valve|bulb B1 is provided in the inlet_port|entrance location to the target box 22 at the line L1.
Moreover, the pressure gauge P1 and valve|bulb B2 are provided in the location which came out of the target box 22 to the line L2.

[0022] Furthermore, in the cyclotron chamber C, while introducing RI gas after the purity measurement returned from the hot lab chamber H through the line L4, RI gas after the administration returned from the PET laboratory R is introduced through the line L6,
The gas collection|recovery apparatus 26 which collect|recovers these RI gases is provided.

[0023] In the hot lab chamber H, the line L3 connects to the automatic-synthesis apparatus 23, and the purity measuring apparatus 24 which measures the purity of RI gas sent from the automatic-synthesis apparatus 23 is provided.
The pressure gauge P2 is provided in this purity measuring apparatus 24.
The purity measuring apparatus 24 sends out RI gas after a purity measurement to the PET laboratory R through the line L5.
Moreover, the purity measuring apparatus 24 returns RI gas after a purity measurement to the cyclotron chamber C through the line L4.

[0024] The PET laboratory R has the attic|under-roof space (2nd division) T which is the space which the test subject storage chamber S was partitioned off by the test subject storage chamber (1st division) S and the ceiling 12 which are the space in which test subject SB is accommodated, and was formed above the test subject storage chamber S.

[0025] In the test subject storage chamber S of the PET laboratory R, the line L5 connects to the purity measuring apparatus 24, and RI gas sent from the purity measuring apparatus 24 is introduced,
While the gas suction|inhalation apparatus 20 for making test subject SB suck|inhale the said RI gas through the line L7 and the body of test subject SB mount, the PET apparatus 10 for imaging

the inside of the body of test subject SB which suck|inhaled RI gas from the gas suction|inhalation apparatus 20, and carrying out a PET test|inspection is installed.
Here, it attaches to the opening|mouth of test subject SB, and the mask 11 for supplying RI gas to test subject SB smoothly is provided at the front-end|tip of the line L7.

[**0026**] The gas suction|inhalation apparatus 20 collect|recovers the expiration|expired_air of test subject SB containing RI gas after administration through the mask 11 and the line L7 while supplying RI gas to test subject SB through the mask 11.
Furthermore, the gas suction|inhalation apparatus 20 is returned to the gas collection|recovery apparatus 26 which mentioned above the expiration|expired_air of collect|recovered test subject SB through the line L6.

[**0027**] Furthermore, in the attic|under-roof space T of the PET laboratory R, while attracting|sucking the air (gas) A in the test subject storage chamber S, the exhaust apparatus 18 for exhausting to the exterior (outdoor) of the PET laboratory R is provided.

[**0028**] The ceiling filter 13 by which this exhaust apparatus 18 was installed in the opening part of the ceiling 12,
The line L7 which is a duct which introduces the air A which passed the ceiling filter 13,
While delaying outdoor discharge|emission of the air A, the air A is attracted|sucked from the test subject storage chamber S with the filter 16 for removing a part of RI gas contained in the air A which passed through the exhaust_gas|exhaustion RI density|concentration reduction apparatus 15 for reducing the density|concentration of RI gas which may be contained in the air A, and the exhaust_gas|exhaustion RI density|concentration reduction apparatus 15,
It has the ventilating fan 17 discharged|emitted to outdoor through the line L8 by making the attracted|sucked air A into exhausted air.

[**0029**] FIG. 2 is a schematic sectional drawing which shows the exhaust_gas|exhaustion RI density|concentration reduction apparatus 15 in FIG. 1.
It is for providing the exhaust_gas|exhaustion RI density|concentration reduction apparatus 15 shown in FIG.1 and FIG.2 in the laboratory in which test subject SB to which RI gas for a PET test|inspection (radioactive gas) was administered is accommodated, and reducing RI (<15>O) density|concentration in exhausted air,
The delay section (discharge|emission delay means) 14 which delays outdoor discharge|emission of the air A in the test subject storage chamber S is provided.
In addition, the exhaust_gas|exhaustion RI density|concentration reduction apparatus 15 may be suitably provided with the damper for gas|air volume adjustment (not shown), etc. other than the delay section 14.

[**0030**] This delay section 14 has the duct part (tubular part) 31 of cross-sectional rectangular shape which lets the air A pass.
This duct part 31 has the inlet_port|entrance part 33 to which the line L7 is connected, and the exit part 34 connected to the filter 16.
Furthermore, the duct part 31 has two or more (FIG. 2 four pieces) curved parts 32 for changing the direction to which the duct part 31 extends.
In more detail, 180 degrees of direction to which the duct part 31 extends has changed to the horizontal direction by making the curved part 32 into an inflection point.
That is, the flow path F1 formed in the duct part 31 is made into the shape from which 180 degrees of flows of the air A change to a horizontal direction before and behind the curved part 32.

[**0031**] Length (The length of the flow path F1 from the inlet_port|entrance part 33 to the exit part 34) of the duct part 31 can be suitably determined according to the half life of the amount of exhaust air by the ventilating fan 17, RI gas concentration which may be contained in the air A, and its RI gas, the emission-standard value of RI, etc.
For example, if a cross section provides the duct part 31 which has a magnitude|size which is 1 m * 1 m covering length of 33 m when the amount of exhaust air is 500 m<3>/h, time for the air A to retain in the delay section 14 will be 4 minutes.
Therefore, when RI gas which consists of <15>O is used, since the half life is about 2 minutes, the residence time of the air A in the delay section 14 will be about 2 time the half life.

[**0032**] In addition, the number of the curved parts 32 may be 1-3, and 5 or more may be sufficient as them.
Moreover, 90 degrees of direction to which the duct part 31 extends may change by making the curved part 32 into an inflection point, and may change 45 degrees.
Furthermore, the direction to which the duct part 31 extends may change to an up-down direction.
Thus, the duct part 31 can be suitably construct|laid in the suitable shape decided with the shape in the attic|under-roof space T, etc.
Moreover, the duct part 31 is not limited to when it is cross-sectional rectangular shape,
A cross-section circular shaped circular pipe may be sufficient.

[**0033**] According to the exhaust apparatus 18 which has such an exhaust_gas|exhaustion RI density| concentration reduction apparatus 15
The air A in the test subject storage chamber S is attracted|sucked through the line L7 by the action|operation of the ventilating fan 17, and the attracted|sucked air A is introduced in the delay section 14.
The air A introduced into the delay section 14 passes through the inside of the flow path F1 over predetermined time, without the flow stopping.
And the air A which came out of the delay section 14 is discharged|emitted from the discharge side of the ventilating fan 17 in the exterior of the PET laboratory R, after a part of RI gas is removed by the filter 16.

[**0034**] Here, if some clearance gaps exist between the mask 11 attached to the opening|mouth of test subject SB, and the face of test subject SB when a PET test|inspection is carried out, RI gas may leak out from this clearance gap.
RI gas leaked out in this way spread|diffuses the inside of the test subject storage chamber S, becomes a predetermined density|concentration, and is contained in the air A.
And if this air A is attracted|sucked by the exhaust apparatus 18, while passing the delay section 14, the radioactivity of RI gas contained in the air A is attenuate|damped below to a predetermined management value.

[**0035**] For example, when the residence time in the delay section 14 is about 2 time the half life like the example mentioned above, the density|concentration of RI contained in the air A can be reduced to about 1/4 in the inlet_port|entrance part 33 and the exit part 34 of the delay section 14.

[**0036**] Since the filter 16 and the ventilating fan 17 were only provided in the line L7 in the conventional exhaust apparatus, the length of the flow path through which the air A flows was as short as about 10m.
Therefore, when RI gas was contained in the air A, there existed a possibility of discharging| emitting while the air A short-circuits to outdoor from the inside of the PET laboratory R and RI density|concentration is not reduced.
Moreover, if discharge|emission of the air A is delayed by restrict|squeezing the ventilation gas|air volume by the ventilating fan 17, RI gas will retain in the PET laboratory R,
There existed a possibility that the exposure to the medical staff who operate|works in the PET laboratory R might arise.

[0037] According to the exhaust_gas|exhaustion RI density|concentration reduction apparatus 15 of this embodiment
Since outdoor discharge|emission of the air A is delayed by the delay section 14 even if it is a case where RI gas leaks out in the test subject storage chamber S, by this delay, the radioactivity of RI gas can be attenuated and RI density|concentration in exhausted air can be reduced desired.
Moreover, since the flow of the air A is not stopped|fastened, it is not necessary to provide the space for retention of gas, and the air A can be discharged|emitted continuously.
Moreover, since the exhausted air which the radioactivity attenuate|damped below to the management value is discharged|emitted to the area outside a radiation management division, continuously different test subject SB can be test|inspected and the frequency|count of a test|inspection can be increased.

[0038] Moreover, since the duct part 31 of the delay section 14 has the curved part 32 for changing the direction to which the duct part 31 extends, few installation spaces are used effectively,
A longer pipe line (flow path F1) can be provided, and the reduction effect of RI density|concentration by the above-mentioned radioactive attenuation|damping can be exhibited suitably.

[0039] Moreover, it can prevent that the delay section 14 exposes it to the test subject storage chamber S in which test subject SB is accommodated since the delay section 14 is arrange|positioned in the attic|under-roof space T, and the improvement of an external appearance is achieved.

[0040] (2nd Embodiment)
FIG. 3 is a schematic sectional drawing which shows the exhaust_gas|exhaustion RI density|concentration reduction apparatus concerning 2nd Embodiment.
A different point from the exhaust_gas|exhaustion RI density|concentration reduction apparatus 15 of 1st Embodiment which exhaust_gas|exhaustion RI density|concentration reduction apparatus 15B of this embodiment shown in FIG. 3 showed in FIG. 2,
It is the point provided with the delay section (discharge|emission delay means) 50 which comprises the chamber 55 which replaces with the delay section 14 which it has, and has the predetermined volume duct part 31.

[0041] This chamber 55 is a hollow container which comprises a substantially rectangular parallelepiped shape, and has the inlet_port|entrance part 53 to which the line L7 (refer FIG. 1) is connected, and the exit part 54 connected to the filter 16 (refer FIG. 1).
Moreover, the chamber 55 has predetermined length, the inlet_port|entrance 53 is provided in one edge part, and the exit 54 is provided in the other edge part.

[0042] Furthermore, it is the plate-shaped member perpendicularly|vertically standingly arranged by the 1st wall part (wall part of illustration lower side) 56 which is a wall part of 1 which forms the

chamber 55 towards the 2nd wall part (wall part of an illustration upper side) 57 which opposes the 1st wall part 56,
Comprising:
The several partition plate (2nd partition plate) 51 mutually spaced apart predetermined length in a direction perpendicular|vertical to the plate|board surface is arranged in parallel.
It is spaced apart by a predetermined distance of the front-end|tip part (edge part of an illustration upper side) 51a of this partition plate 51 from the 2nd wall part 57.
Moreover, it is the plate-shaped member perpendicularly|vertically standingly arranged by the 2nd wall part 57 towards the 1st wall part 56,
Comprising:
It arrange|positions so that the partition plate 51 may be faced from a direction perpendicular|vertical to the plate|board surface, and the several partition plate (2nd partition plate) 52 mutually spaced apart predetermined length in the said perpendicular|vertical direction is arranged in parallel.
It is spaced apart by a predetermined distance of the front-end|tip part (edge part of illustration lower side) 52a of this partition plate 52 from the 1st wall part 56.

[**0043**] The flow path F2 formed in the chamber 55 which has such a structure is made into the shape which changes the flow of the air A to the downflow from upward flow, or upward flow from a downflow by the partition plates 51 and 52 alternately standingly arranged from the 1st wall part 56 and the 2nd wall part 57.

[**0044**] According to exhaust_gas|exhaustion RI density|concentration reduction apparatus 15B of this embodiment
It prevents that the air A short-circuits and flows into the chamber 55 with the partition plates 51 and 52 provided in the chamber 55,
Discharge|emission of the air A can be reliably delayed by lengthening the flow path F2.

[**0045**] (3rd Embodiment)
FIG. 4 is a front view which shows schematic structure of the exhaust_gas|exhaustion RI density|concentration reduction apparatus concerning 3rd Embodiment.
With the exhaust_gas|exhaustion RI density|concentration reduction apparatus 15 of 1st and 2nd embodiment, and 15B, it is different of the exhaust_gas|exhaustion RI density|concentration

reduction apparatus 15C of this embodiment, and it is arrange|positioned in the test subject storage chamber S of the PET laboratory R.
Moreover, the discharge port 66 for discharging|emitting the air A from the test subject storage chamber S in the corner of the ceiling 12 is formed in the ceiling 12 of test subject storage chamber S upper direction,
The line L8 is connected to this discharge port 66.
The line L8 is provided with the filter 16 and the ventilating fan 17 which were shown outside the PET laboratory R at FIG. 1.
In addition, in FIG. 4, the attic|under-roof space T is abbreviate|omitting illustration.

[**0046**] The exhaust_gas|exhaustion RI density|concentration reduction apparatus 15C has the partition plate (1st partition plate) 61 which was provided so that it might suspend down towards the floor surface 65 from the ceiling 12 between the PET apparatus 10 and the discharge port 66 and which has a predetermined width.
It is spaced apart by a predetermined distance of the lower end part (edge part of illustration lower side) 61a of the partition plate 61 from the floor surface 19 of the test subject storage chamber S.
Moreover, the right-and-left end parts (edge part of an illustration near side and a back|inner side) of the partition plate 61 may be contact|abutted on the side wall of the test subject storage chamber S, and may be spaced apart from the said side wall.

[**0047**] This partition plate 61 is for changing the flow direction F3 of the air A.
That is, the partition plate 61 prevents that the air A short-circuits to the discharge port 66, and flows from the PET apparatus 10,
Outdoor discharge|emission of the air A is delayed by changing the flow direction F3.
In the exhaust_gas|exhaustion RI density|concentration reduction apparatus 15C, such a partition plate 61 is provided and the delay section (discharge|emission delay means) 60 is comprised.

[**0048**] According to the exhaust_gas|exhaustion RI density|concentration reduction apparatus 15C of this embodiment
Since the flow direction F3 of the air A is changeable with the partition plate 61, discharge|emission of the air A can be reliably delayed in the limited space in the test subject storage chamber S, and the reduction effect of RI density|concentration by radioactive attenuation|damping can be exhibited suitably.
Moreover, the said effect can be exhibited by simple structure, without requiring complicated components.

[**0049**] As mentioned above, although embodiment of this invention was described, this invention is not limited to the said embodiment.
For example, the said 1st and 2nd embodiment demonstrated the case where the filter 16 and the ventilating fan 17 of the exhaust apparatus 18 were provided in the attic|under-roof space T, However,
The filter 16 and the ventilating fan 17 may be provided in outdoor [of the PET laboratory R].
Moreover, the delay section 14 of the exhaust_gas|exhaustion RI density|concentration reduction apparatus 15 and the delay section 50 of exhaust_gas|exhaustion RI density|concentration reduction apparatus 15B may be installed in the test subject storage chamber S.
Moreover, you may combine suitably the exhaust_gas|exhaustion RI density|concentration reduction apparatus 15, and exhaust_gas|exhaustion RI density|concentration reduction apparatus 15B and the exhaust_gas|exhaustion RI density|concentration reduction apparatus 15C.

DESCRIPTION OF SYMBOLS

[**0050**] 10... PET apparatus,
14... Delay section (discharge|emission delay means),
15, 15B, 15C... Exhaust_gas|exhaustion RI density|concentration reduction apparatus,
31... Duct part (tubular part),
32... Curved part,
50... Delay section (discharge|emission delay means),
51,52... Partition plate (2nd partition plate),
55... Chamber,
60... Delay section (discharge|emission delay means),
61... Partition plate (1st partition plate),
A... Air (gas),
R... PET laboratory (laboratory),
S... Test subject storage chamber (1st division),
SB... Test subject,
T... Attic|under-roof space (2nd division).

It is provided in the laboratory in which the PET apparatus was installed, and a discharge|emission delay means to delay the discharge|emission to said outdoor of the said gas is provided, without stopping|fastening the flow of the gas discharged|emitted from the inside of the said laboratory outdoor, The exhaust_gas|exhaustion RI density|concentration reduction apparatus characterized by the above-mentioned.

The said discharge|emission delay means has a tubular part which lets the said gas pass,
The said tubular part has a curved part for changing the direction to which the said tubular part extends,
The exhaust_gas|exhaustion RI density|concentration reduction apparatus of Claim 1 characterized by the above-mentioned.

The said discharge|emission delay means has a 1st partition plate for changing the flow direction of the said gas,
The exhaust_gas|exhaustion RI density|concentration reduction apparatus of Claim 1 or 2 characterized by the above-mentioned.

The said discharge|emission delay means comprises the chamber which has the predetermined volume,
The 2nd partition plate is provided in the said chamber,
The exhaust_gas|exhaustion RI density|concentration reduction apparatus of the any one of Claims 1-3 description characterized by the above-mentioned.

The said laboratory has the 2nd division whose 1st division which is space in which a test subject is accommodated and said 1st division are the partitioned-off space,
The said discharge|emission delay means is arrange|positioned in said 2nd division,
The exhaust_gas|exhaustion RI density|concentration reduction apparatus of the any one of Claims 1-4 description characterized by the above-mentioned.